MathDB 367324: exact boundary values for all nonmultiples of three

Principia Math In Progress

18 September 2026

Task 1: Complete corrected classification

Let $M(z) = 1 + zM(z) + z^2M(z)^2$ and put

$$H_N^{(r)} = \det([z^{i+j}]M(z)^r)_{0\le i,j<N}, \qquad H_0^{(r)} = 1.$$

Wang and Zhang's Conjecture 7 (arXiv:2502.21050, p. 4) contains sign errors. In particular, its proposed nonzero sum next to a pair of zero determinants is actually zero. We determine both summands individually, as well as every other family singled out by the conjecture, for all exponents and block indices.

Theorem 1. *Let $r \ge 2$, $3 \nmid r$, and $n \ge 0$. Set*

$$L = 3rn, \qquad \epsilon_r = (-1)^{\binom{r}{2}}, \qquad \eta_r = \begin{cases} 1, & r \equiv 1 \pmod 3, \\ (-1)^r, & r \equiv 2 \pmod 3. \end{cases}$$

Then

$$H_L^{(r)} = H_{L+1}^{(r)} = \epsilon_r^n, \tag{1}$$

$$H_{L+r}^{(r)} = H_{L+r+1}^{(r)} = (-1)^{r-1}\epsilon_r^{n+1}, \tag{2}$$

$$H_{L+2r}^{(r)} = H_{L+2r+1}^{(r)} = 0, \tag{3}$$

$$H_{L+2r-1}^{(r)} = -H_{L+2r+2}^{(r)} = \eta_r\epsilon_r^n[2r(n+1)]^{r-2}, \tag{4}$$

$$H_{L-1}^{(r)} + H_{L+2}^{(r)} = -\frac{r(r-3)}{2}\epsilon_r^n. \tag{5}$$

For $n = 0$, (5) uses $H_{-1}^{(r)} = 0$; otherwise take $n \ge 1$. For the remaining exponent $r = 1$, $H_N^{(1)} = 1$ for all $N \ge 0$.

Thus the source's claimed magnitude $[2r(n+1)]^{r-2}$ is the magnitude of *each* of the two neighbors in (4), whose signs are opposite. For example, $r = 4, n = 0$ gives $H_7^{(4)} = 64$ and $H_{10}^{(4)} = -64$. Their sum is zero, not ± 64. Also $H_0^{(4)} = 1$ and $H_4^{(4)} = -1$, so the single common sign printed for all four nonzero boundary values is not valid. The proof below establishes the complete formulas; the result does not rely merely on pointing out these discrepancies.

The weighted boundary identities below were established in the companion solution to MathDB 367323. We reproduce their proof and then evaluate the additional minor at the zero boundaries to obtain the individual values in (4). For a real parameter a, let $M_a = 1 + azM_a + z^2M_a^2$, and write its convolution determinants as $H_N^{(r)}(a)$. Define

$$P_j(t) = U_j((t-a)/2), \qquad A(t) = P_{r-1}(t), \qquad B_k(t) = U_k(T_r((t-a)/2))\,.$$

Here $T_j(\cos\theta) = \cos(j\theta)$ and $U_j(\cos\theta) = \sin((j+1)\theta)/\sin\theta$; P_j and B_k are monic, of degrees j and rk.

Theorem 2 (Weighted boundary identities)**.** *For $r \geq 2$, $k \geq 0$, $N = rk$, and every real or complex a,*

$$H_N^{(r)}(a) = H_{N+1}^{(r)}(a) = \epsilon_r^k B_k(0)^{r-1}, \tag{6}$$

$$H_{N-1}^{(r)}(a) + H_{N+2}^{(r)}(a) = \left(r - \binom{r}{2} a^2\right) H_N^{(r)}(a). \tag{7}$$

The convolution functional and its determinant

For real a, let μ_a be the translated semicircle measure

$$d\mu_a(t) = \frac{\sqrt{4-(t-a)^2}}{2\pi}\, dt \quad (a-2 \leq t \leq a+2).$$

Its moments are $[z^j]M_a(z)$, and its monic orthogonal polynomials are P_j, with $\int P_i P_j \, d\mu_a = \delta_{ij}$. Both statements follow by $t = a + 2\cos\theta$: the centered moments are the Catalan numbers and the orthogonality integral is the sine integral. Set $Q(t) = t^{r-1}A(t)$ and $L(f) = \int f(t)Q(t)\, d\mu_a(t)$. Then

$$[z^j]M_a(z)^r = L(t^j). \tag{8}$$

Indeed $G(w) = w^{-1}M_a(w^{-1})$ satisfies $G^2 = (w-a)G - 1$, so induction gives $G^r = P_{r-1}(w)G - P_{r-2}(w)$. Multiplication by w^{r-1} and extraction of w^{-j-1} gives (8). The functional L need not be positive.

We use the following classical Christoffel identity in an algebraic form. Its evaluation form is Theorem 1 of Krattenthaler, arXiv:2101.04225, with repeated roots covered by Proposition 5. A proof is included to make the argument self-contained.

Lemma 1. *If Q is monic of even degree d, the determinant of the moments of $Q\, d\mu_a$ of order m is the determinant of the $d \times d$ matrix whose jth column is the coefficient vector of $P_{m+j} \bmod Q$, for $0 \leq j < d$.*

Proof. First suppose Q has distinct roots $\alpha_1, \ldots, \alpha_d$. Expansion of determinants gives the moment determinant as

$$\frac{1}{m!} \int \cdots \int \Delta(x)^2 \prod_{i=1}^{m} Q(x_i) \prod_{i=1}^{m} d\mu_a(x_i).$$

The mixed Vandermonde satisfies $\Delta(x,\alpha) = \Delta(x)\Delta(\alpha)\prod_{i,j}(\alpha_j - x_i)$. Replace monomials in both Vandermondes by the monic P_j. Orthogonality leaves only columns $0, \ldots, m-1$ for the x rows, with integral $m!$. Hence the determinant is

$$(-1)^{md} \frac{\det(P_{m+j}(\alpha_i))_{1\leq i\leq d,\, 0\leq j<d}}{\det(\alpha_i^j)_{1\leq i\leq d,\, 0\leq j<d}}.$$

The sign is one. Evaluation of remainders factors the numerator matrix as the denominator matrix times the coefficient matrix. The identity extends to repeated roots because both sides are polynomials in the coefficients of the monic Q; polynomial division introduces no denominators. It also includes $m = 0$. □

Exact evaluation at a block boundary

Put $N = rk$ and $c = N + r - 1$. The sine formula gives

$$P_c = AB_k. \tag{9}$$

In the remainder determinant of Lemma 1, retain its first $r-1$ columns $P_N, \ldots, P_{N+r-2}$. For each $1 \leq j \leq r-2$, add column P_{c-j} to column P_{c+j}. The last $r-1$ columns then become

$$AB_kS_0, \ldots, AB_kS_{r-2}, \qquad S_0 = 1, \quad S_j(t) = 2T_j((t-a)/2) \ (j \geq 1).$$

Here $P_{c+j} + P_{c-j} = S_jP_c$, and each S_j is monic of degree j.

Use the following monic basis of $\mathbb{C}[t]/(t^{r-1}A)$:

$$P_0, \ldots, P_{r-2}, \quad AS_0, \ldots, AS_{r-2}.$$

Its change-of-basis determinant from the monomials is one. Reduction modulo A gives an upper block with the first $r-1$ columns and zero in the last columns. At the simple roots $\rho_j = a + 2\cos(j\pi/r)$ of A,

$$P_{N+h}(\rho_j) = (-1)^{kj} P_h(\rho_j) \qquad (0 \le h \le r-2).$$

Thus the upper block determinant is $\prod_{j=1}^{r-1}(-1)^{kj} = \epsilon_r^k$. The lower right block represents multiplication by B_k in $\mathbb{C}[t]/(t^{r-1})$. In its monomial basis this matrix is triangular, with diagonal $B_k(0)$, so its determinant is $B_k(0)^{r-1}$. This proves the formula for H_N in (6). No coprimality of A and t was assumed.

The adjacent determinant and the paired sum

Temporarily choose a real a for which $B_k(0) \neq 0$. This excludes only finitely many values. Equation (9) gives

$$L(t^j B_k) = \int t^{j+r-1} P_{N+r-1}(t)\, d\mu_a(t) = 0 \qquad (0 \le j < N).$$

Consequently $q_N = B_k$ is the unique monic polynomial of degree N orthogonal to lower degrees for L, since $H_N \neq 0$. The elementary identity $U_k(x)^2 = \sum_{h=0}^{k} U_{2h}(x)$ yields

$$AB_k^2 = \sum_{h=0}^{k} P_{(2h+1)r-1}. \tag{10}$$

For $j = 0, 1, 2$, every $h \ge 1$ term is orthogonal to t^{r-1+j}, because $r \ge 2$. Therefore

$$L(q_N^2) = 1, \qquad L(tq_N^2) = ra, \qquad L(t^2 q_N^2) = r + \binom{r+1}{2} a^2. \tag{11}$$

For completeness, the last two values are the coefficients of P_{r-1} in t^r and t^{r+1}. The recurrence $tP_j = P_{j+1} + aP_j + P_{j-1}$, with $P_{-1} = 0$, gives respectively ra and $\binom{r+1}{2}a^2 + r$: in the latter coefficient there are two level steps or one down step among the otherwise up steps.

The Gram determinant identity $L(q_N^2) = H_{N+1}/H_N$ now proves $H_{N+1} = H_N$. In particular q_{N+1} also exists. Suppose first that $N \ge 1$. Let $G_N = (L(t^{i+j}))_{0 \le i,j < N}$ and write

$$tq_N = q_{N+1} + ra\, q_N + v, \qquad \deg v < N.$$

For $j < N-1$, $L(t^j v) = 0$, whereas $L(t^{N-1}v) = 1$. Thus the coefficient vector of v is $G_N^{-1} e_{N-1}$, and

$$L(v^2) = (G_N^{-1})_{N-1,N-1} = \frac{H_{N-1}}{H_N}.$$

The three summands in the displayed decomposition are mutually orthogonal. Taking squared norms and using (11) gives

$$r + \binom{r+1}{2} a^2 = \frac{H_{N+2}}{H_{N+1}} + r^2 a^2 + \frac{H_{N-1}}{H_N},$$

which is (7). This argument never divides by H_{N-1} or H_{N+2}, which are allowed to vanish. For $N = 0$, the same identity is directly $H_2 = L(t^2) - L(t)^2 = r - \binom{r}{2}a^2$. Finally, the asserted identities are polynomial identities in a, so they extend from the generic real values to every complex value, including $B_k(0) = 0$. This proves Theorem 2.

The missing minor at a zero boundary

Now put $a = 1$, $k = 3n+2$, and $N = rk$, with $3 \nmid r$. Write $B = B_k$ and $D = B_{k-1}$. Since $T_r(-1/2) = -1/2$,

$$B(0) = 0, \qquad D(0) = -1, \qquad b := B'(0) = \begin{cases} -2r(n+1), & r \equiv 1 \pmod 3, \\ 2r(n+1), & r \equiv 2 \pmod 3. \end{cases} \tag{12}$$

To verify the derivative, set $t = 1 + 2\cos\theta$. Then

$$B(t) = \frac{\sin((k+1)r\theta)}{\sin(r\theta)}, \qquad \theta\big|_{t=0} = \frac{2\pi}{3}, \qquad \left.\frac{d\theta}{dt}\right|_{t=0} = -\frac{1}{\sqrt{3}}.$$

At this point the numerator is zero and its cosine is one, while $\sin(2\pi r/3)$ equals $\sqrt{3}/2$ or $-\sqrt{3}/2$. This proves (12) and, in particular, $b \neq 0$.

Apply Lemma 1 to $H_{N-1}^{(r)}$. Its columns are

$$P_{N-1}, P_N, \ldots, P_{N+2r-4}.$$

Move the first column past the next $r-1$ columns, contributing the sign $(-1)^{r-1}$. As before, use $P_{c+j}+P_{c-j} = S_j P_c$, now for $1 \le j \le r-3$, where $c = N+r-1$. Since $P_{N-1} = AD$, the reordered, transformed columns are

$$P_N, \ldots, P_{N+r-2}, \quad AD, \quad ABS_0, \ldots, ABS_{r-3}.$$

For $r = 2$ the last list and the corresponding column additions are empty. In the same monic quotient basis used above, the upper block again has determinant ϵ_r^k. The lower block is represented modulo t^{r-1} by

$$D, \quad BS_0, \ldots, BS_{r-3}.$$

Replacing the monic S_j by monomials uses column operations of determinant one. Because $B(t) = bt + O(t^2)$, the resulting matrix is triangular with diagonal $D(0), b, \ldots, b$. Hence

$$H_{N-1}^{(r)} = (-1)^{r-1}\epsilon_r^k D(0) b^{r-2} = (-1)^r \epsilon_r^k b^{r-2}. \tag{13}$$

Since $H_N = H_{N+1} = 0$, the polynomial identity (7) gives $H_{N+2} = -H_{N-1}$, with no division by these zero determinants. For $r \equiv 1 \pmod 3$, the signs from $(-1)^r$ and b^{r-2} cancel. For $r \equiv 2 \pmod 3$, they give $(-1)^r$. Also $\epsilon_r^k = \epsilon_r^n$. These observations prove (4) with exactly the stated η_r.

The other boundary values

At $a = 1$ and $3 \nmid r$, the sequence $B_k(0) = U_k(-1/2)$ repeats $1, -1, 0$ with period three. Substitute $k = 3n, 3n+1, 3n+2$ in (6) to obtain (1)–(3). Substitution of $k = 3n$ into (7) gives (5). Finally, when $r = 1$, the original semicircle moments have monic orthogonal polynomials of squared norm one, so every Hankel determinant is one. This completes the classification and deals with that separately known exponent explicitly.

Task 2: Closed formulas inside each block

Find direct formulas for the residue classes not listed in Theorem 1, refining the general polynomial-block theorem.

Task 3: Rank and nullspaces at the zero boundaries

Determine the exact ranks and explicit nullspace bases of the two singular Hankel matrices at indices $3rn+2r$ and $3rn+2r+1$.

Task 4: Least possible block lengths

Determine the smallest block length for which all residue classes are polynomial up to a parity factor, and classify exceptional exponents.

Task 5: Combinatorial cancellation and boundary signs

Construct an involution on path families that explains both the two successive zeros and the opposite neighboring signs in (4).